1. Clinical Question & Core Hypothesis

Clinical Question

S100A9-driven inflammation should promote immunogenic cell death and recruit CD8+ T cells that limit invasion, yet some S100A9/TIL-rich bladder tumors still progress. Why does local inflammation and lymphocyte presence fail to suppress invasion in these cases? 2) Tier 1 paths used: Core Path 1 (S100A9 → cell death → CD8+ T cells → protect against invasion) provides the immune/inflammatory axis; Core Path 3 (PFKFB3 → cancer stemness) provides a tumor-intrinsic metabolic/stemness axis that can antagonize immune control. I built the hypothesis on the intersection of these two mechanistic backbones because they directly address the failure of inflammation to restrain invasion. 3) Tier 2 support: LYPD1 (actionable surface target) and negative association between PFKFB3 and stroma content support a microenvironmental and targetable component; EPB41L2 and cross-cancer evidence for PFKFB3 (HCC prognostic relevance) further justify testing PFKFB3 as a conserved regulator of aggressive behavior. 4) Endpoint choice: Progression-free interval (PFI) was chosen because the core question concerns invasion and early progression rather than long-term survival or specific therapy response; PFI best captures clinically relevant disease progression tied to invasion. 5) Deliberate exclusions: LYPD1, EPB41L2, and stromal features were placed in secondary_modifiers rather than the core hypothesis to keep the mechanistic question sharp (max 2 anchors). I also avoided introducing gene proxies (e.g., CD8A) that were not explicitly provided by the Ontologist?

Working Hypothesis

Do TIL-dense urothelial bladder tumors that express S100A9 and exhibit high tumor-intrinsic PFKFB3-driven glycolysis/stemness fail to restrain invasion because metabolic reprogramming decouples S100A9-induced immunogenic cell death from effective CD8+ cytotoxicity? If this is true, pharmacologic or genetic inhibition of PFKFB3 in these tumors should restore CD8+ T cell–mediated cytotoxicity and reduce invasion.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	S100A9, PFKFB3
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

S100A9: S100A9 → cell death → CD8+ T cells → protect against invasion) provides the immune/inflammatory axis; Core Path 3 (PFKFB3 → cancer stemness) provides a tumor-intrinsic metabolic/stemness axis that can antagonize immune control.

PFKFB3: PFKFB3 and stroma content support a microenvironmental and targetable component; EPB41L2 and cross-cancer evidence for PFKFB3 (HCC prognostic relevance) further justify testing PFKFB3 as a conserved regulator of aggressive behavior.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	LYPD1	Supporting modifier from knowledge graph
Secondary	EPB41L2	Supporting modifier from knowledge graph
Secondary	stroma content	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: that can antagonize immune control. Tier 2 signals (LYPD1, EPB41L2, stromal context) offer context for modulating microenvironment and potential therapeutic targets.

Prediction 2: and measure immunogenic cell death markers (calreticulin exposure, HMGB1 release) and glycolytic flux (Seahorse). - In vivo orthotopic bladder cancer models with manipulated PFKFB3 expression or pharmacologic inhibition to evaluate effects on CD8+ T cell function, invasion markers, and microenvironmental metabolism; use histology and flow cytometry to quantify outcomes.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- LYPD1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EPB41L2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.